

Course Projects for Deep Learning Course

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- 1) Speaker verification on edge devices (Jagabandhu)
- 2) Deep learning approaches for voice activity detection (Rishith)
- 3) Periodic-aperiodic decomposition of speech using deep learning (Rishith)
- 4) Speech recognition system for intelligent tutoring system (Prasanna)
- 5) Dialog system for intelligent tutoring system (Prasanna)
- 6) Speech synthesis system for intelligent tutoring system (Prasanna)
- 7) Speech enhancement using deep learning (Rishith)
- 8) Time-delay estimation using deep learning (Rishith)
- 9) Spoof detection using deep learning (Rishith)
- 10) End2End Language change detection (code-switch utterance with no prior language information) (Jagabandhu)
 - a. Extending our earlier work on E2E Diarization (<https://ieeexplore.ieee.org/document/10596692>) to train a universal E2E language change detector. The embeddings from the same can be used for efficiently doing language diarization using the change point approach in the DISPLACE corpus, where the utterances have no prior language information.
- 11) Cleft lip and cleft palate (CLP) speech: severity assessment using Modulation based features: (Sishir)
 - a. The initial study suggests that continuity in speech distorts with severity of CLP. With keeping limited amount of data in mind, the guided feature (complex FDLP) design might be beneficial over using standard mel-filter bank and spectrogram based features.
- 12) CLP: severity assessment using features from foundation models: (Sishir)
 - a. Recent literature claims the speech foundation models captures speech dynamics and successfully used in many downstream tasks It might be interesting to see and benchmark the performance of such features () (Whisper, WavLM, Wav2Vec etc.) CLP severity assessment.
- 13) CLP: Severity assessment using explicit grapheme representation based features. (Sishir)
 - a. The aim here is to observe how the grapheme posterior (GP) dynamics change with the severity of CLP. The pre-trained language-specific models like W2V conformer, Whisper, and MMS models can be used to obtain the GP.

14) Dementia detection and cognitive decline prediction using acoustic features
(Sishir) (<https://processchallenge.github.io/schedule/>)

15) Dementia detection and cognitive decline prediction using combined acoustic and linguistic features. (Sishir)